# Notes

## Introduction to Quadratic Equations

### Welcome and Session Overview

This session continues from Module 2, focusing on quadratic equations. We will build upon the foundational concepts discussed previously.

### Recap of Previous Session

In the last session, we covered:

- What a polynomial is.
- The definition of a quadratic equation.
- The relationship between roots and coefficients of a quadratic equation.

These topics are fundamental and are often introduced in earlier classes, forming the basis for more advanced concepts in this module.

## Recap: Sum and Product of Roots

### Understanding S1 and S2

In the context of quadratic equations, we use specific notations for the sum and product of roots:

- **S1 (Sum of roots taking 1 at a time):** This refers to the sum of the roots. For a quadratic equation $ax^2 + bx + c = 0$, the sum of the roots is given by $S1 = -b/a$.
- **S2 (Sum of roots taking 2 at a time):** This refers to the product of the roots. For a quadratic equation $ax^2 + bx + c = 0$, the product of the roots is given by $S2 = c/a$.

### Quiz Problem: Finding the Sum of Roots

**Question:** If the product of the roots of the equation $2x^2 + (a + 1)x + 4 = 0$ is 2, then the sum of the roots is:

**Solution:**

1. **Identify the coefficients:** For the given equation $2x^2 + (a + 1)x + 4 = 0$, we have:
    - $a = 2$
    - $b = (a + 1)$
    - $c = 4$

3. **Solve for 'a':** The problem statement implies that the coefficient 'b' contains an unknown 'a'. Let's re-evaluate the problem statement. It seems there was a slight confusion in the verbal explanation. Let's assume the equation was meant to be $2x^2 + (a + 1)x + 4 = 0$ and the product of roots is 2. The product of roots is $c/a = 4/2 = 2$. This means the given product of roots is consistent with the equation's constant term and leading coefficient. The variable 'a' in the coefficient of 'x' is what we need to find to determine the sum of roots.

   Let's re-interpret the lecturer's steps: "This means $4 = 2 * (a + 1)$. So, $4 = 2a + 2$. $2a = 2$, and $a = 1$." This step seems to be solving for a different 'a' than the coefficient of $x^2$. It implies that the constant term 'c' was somehow related to $2(a + 1)$. However, in the given equation $2x^2 + (a + 1)x + 4 = 0$, the constant term is 4. The coefficient of $x$ is $(a + 1)$.

   Let's assume the lecturer meant the product of roots is $c/a = 2$, and then used a different interpretation for the value of 'c' or 'a' in the equation. If the product of roots is 2, and the equation is $2x^2 + (a + 1)x + 4 = 0$, then $c/a = 4/2 = 2$. This is already satisfied. The 'a' in $(a + 1)$ is what we need to find.

   Let's follow the lecturer's calculation for 'a': If $4 = 2(a + 1)$ (this implies $c = 2imes(extcoefficientofx)$ which is not standard) $4 = 2a + 2$ $2a = 2$ $a = 1$

   *Self-correction based on standard quadratic form:* If the product of roots is 2, and the equation is $2x^2 + (a + 1)x + 4 = 0$, then $c/a = 4/2 = 2$. This condition is already met. The value of 'a' in $(a + 1)$ is not determined by the product of roots being 2. There might be a missing piece of information or a different interpretation of the problem in the transcript.

   However, if we strictly follow the lecturer's steps where they derived $a = 1$ from $4 = 2(a + 1)$, let's proceed with $a = 1$.

4. **Calculate the sum of roots:** The sum of the roots ($S1$) is given by $-b/a$. In our equation, $b = (a + 1)$ and the leading coefficient is 2. Substitute the value $a = 1$ (as derived by the lecturer): $b = (1 + 1) = 2$

   Now, calculate $S1$: $S1 = -b/a = -(a + 1)/2$ Using $a = 1$ (from the lecturer's derivation): $S1 = -(1 + 1)/2 = -2/2 = -1$

   The lecturer states: "But a is 1, so 2 + 3 is 5. So, -5/2 is the answer." This implies a different 'b' value or a different interpretation of 'a'. If the sum of roots is $-5/2$, then $-(a + 1)/2 = -5/2$, which means $a + 1 = 5$, so $a = 4$. This contradicts $a = 1$.

   Given the discrepancy, let's assume the final answer provided by the lecturer, $-5/2$, is correct and work backward or acknowledge the inconsistency. The lecturer explicitly states "-5/2 is the answer."

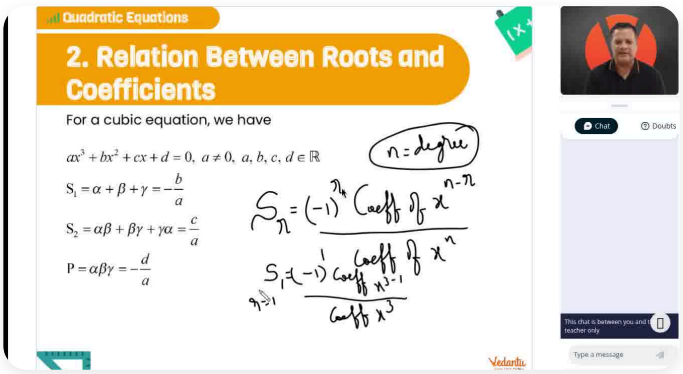
This means the 'a' in the coefficient $(a + 1)$ should be 4 for the sum of roots to be $-5/2$. The lecturer's derivation of $a = 1$ and then stating $-5/2$ as the answer contains a contradiction. For study purposes, it's important to note this inconsistency and understand the correct application of the formulas.

**Correct application of formulas (assuming the question intended for the sum of roots to be $-5/2$):**

- Product of roots: $c/a = 4/2 = 2$ (This condition is met).
- Sum of roots: $S1 = -b/a = -(a + 1)/2$. If the answer is $-5/2$, then $a = 4$.

The key takeaway is the correct formulas for sum and product of roots: $S1 = -b/a$ and $S2 = c/a$.

## Relation Between Roots and Coefficients for Cubic Equations




### General Formula for Sum of Roots

The general formula for the sum of roots $(S_r)$ taking $r$ roots at a time for a polynomial of degree $n$ is given by: $S_r = (-1)^r \frac{\text{Coefficient of } x^{n-r}}{\text{Coefficient of } x^n}$ Here, $n$ represents the degree of the polynomial (the highest power of $x$).

### Cubic Equation Example

For a cubic equation of the form $ax^3 + bx^2 + cx + d = 0$, where $a \neq 0$ and $a, b, c, d \in \mathbb{R}$:

1. **Sum of the roots taking one at a time $(S_1)$:**
   - Here, $n = 3$ and $r = 1$.
   - $S_1 = (-1)^1 \frac{\text{Coefficient of } x^{3-1}}{\text{Coefficient of } x^3} = -\frac{\text{Coefficient of } x^2}{\text{Coefficient of } x^3}$
   - $S_1 = \alpha + \beta + \gamma = -\frac{b}{a}$
2. **Sum of the roots taking two at a time $(S_2)$:**

- $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$

3. **Sum of the roots taking three at a time ($S_3$)**:
   - This is the product of the roots.
   - Here, $n = 3$ and $r = 3$.
   - $S_3 = (-1)^3 \frac{\text{Coefficient of } x^{3-3}}{\text{Coefficient of } x^3} = -\frac{\text{Coefficient of } x^0}{\text{Coefficient of } x^3}$
   - $P = \alpha\beta\gamma = -\frac{d}{a}$

**Summary for Cubic Equation**

For $ax^3 + bx^2 + cx + d = 0$:

- $S_1 = \alpha + \beta + \gamma = -\frac{b}{a}$
- $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$
- $P = \alpha\beta\gamma = -\frac{d}{a}$

## Relation Between Roots and Coefficients for Biquadratic (Fourth Degree) Equations

For a biquadratic equation (degree 4), let the equation be $Ax^4 + Bx^3 + Cx^2 + Dx + E = 0$. Here, $n = 4$.

1. **Sum of the roots taking one at a time ($S_1$)**:
   - $S_1 = (-1)^1 \frac{\text{Coefficient of } x^{4-1}}{\text{Coefficient of } x^4} = -\frac{\text{Coefficient of } x^3}{\text{Coefficient of } x^4}$
   - $S_1 = -\frac{B}{A}$

2. **Sum of the roots taking two at a time ($S_2$)**:
   - $S_2 = (-1)^2 \frac{\text{Coefficient of } x^{4-2}}{\text{Coefficient of } x^4} = +\frac{\text{Coefficient of } x^2}{\text{Coefficient of } x^4}$
   - $S_2 = \frac{C}{A}$

3. **Sum of the roots taking three at a time ($S_3$)**:
   - $S_3 = (-1)^3 \frac{\text{Coefficient of } x^{4-3}}{\text{Coefficient of } x^4} = -\frac{\text{Coefficient of } x^1}{\text{Coefficient of } x^4}$
   - $S_3 = -\frac{D}{A}$

4. **Sum of the roots taking four at a time ($S_4$)**:
   - This is the product of the roots.
   - $S_4 = (-1)^4 \frac{\text{Coefficient of } x^{4-4}}{\text{Coefficient of } x^4} = +\frac{\text{Coefficient of } x^0}{\text{Coefficient of } x^4}$

# Notes

Let's consider a fifth-degree polynomial of the form $Ax^5 + Bx^4 + Cx^3 + Dx^2 + Ex + F = 0$. Here, $n = 5$. Let the roots be $\alpha, \beta, \gamma, \delta, \epsilon$.

1. **Sum of the roots taking one at a time ($S_1$ or $\Sigma\alpha$)**:
   - $S_1 = \alpha + \beta + \gamma + \delta + \epsilon = -\frac{B}{A}$
2. **Sum of the roots taking two at a time ($S_2$ or $\Sigma\alpha\beta$)**:
   - This involves taking all possible products of two roots and summing them up (e.g., $\alpha\beta + \alpha\gamma + \alpha\delta + \alpha\epsilon + \beta\gamma + \ldots$).
   - $S_2 = \frac{C}{A}$
3. **Sum of the roots taking three at a time ($S_3$ or $\Sigma\alpha\beta\gamma$)**:
   - This involves taking all possible products of three roots and summing them up (e.g., $\alpha\beta\gamma + \alpha\beta\delta + \ldots$).
   - $S_3 = -\frac{D}{A}$
4. **Sum of the roots taking four at a time ($S_4$ or $\Sigma\alpha\beta\gamma\delta$)**:
   - This involves taking all possible products of four roots and summing them up.
   - $S_4 = \frac{E}{A}$
5. **Sum of the roots taking five at a time ($S_5$)**:
   - This is the product of all five roots.
   - $S_5 = \alpha\beta\gamma\delta\epsilon = -\frac{F}{A}$

## Relationship Between Zeros/Roots and Coefficients of a Cubic Polynomial

### General Form of a Cubic Polynomial

For a polynomial $P(X) = aX^3 + bX^2 + cX + d$, where $a \neq 0$, the zeros (or roots for an equation) are denoted by $\alpha, \beta, \gamma$.

### Sum of Zeros (S1)

This is the sum of the zeros taken one at a time, also denoted as $\Sigma\alpha$. $S_1 = \alpha + \beta + \gamma = -\frac{b}{a}$

### Sum of Zeros Taken Two at a Time (S2)

This is the sum of the products of the zeros taken two at a time, also denoted as $\Sigma\alpha\beta$. $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = +\frac{c}{a}$

# Notes

### Alternating Signs

Notice the alternating signs in the formulas: $-b/a$, $+c/a$, $-d/a$. This pattern holds true for polynomials of any degree.

**Example: For $3X^3 - 5X^2 - 11X - 3 = 0$**

Here, $a = 3$, $b = -5$, $c = -11$, $d = -3$.

- **S1 (Sum of roots taking one at a time):** $S_1 = -\frac{b}{a} = -\frac{-5}{3} = \frac{5}{3}$
- **S2 (Sum of roots taking two at a time):** $S_2 = +\frac{c}{a} = +\frac{-11}{3} = -\frac{11}{3}$
- **S3 (Product of roots):** $S_3 = -\frac{d}{a} = -\frac{-3}{3} = 1$

## Finding Sum of Roots for a Given Cubic Equation

### Problem Statement

If $\alpha, \beta, \gamma$ are the roots of a cubic equation, find the sum of the roots.

**Example: For $X^3 - 2X + 1 = 0$**

First, identify the coefficients by comparing with the general form $aX^3 + bX^2 + cX + d = 0$.

- Coefficient of $X^3$, $a = 1$
- Coefficient of $X^2$, $b = 0$ (since there is no $X^2$ term)
- Coefficient of $X$, $c = -2$
- Constant term, $d = 1$

### Calculation of S1 (Sum of roots taking one at a time)

The sum of the roots is given by the formula: $S_1 = \alpha + \beta + \gamma = -\frac{b}{a}$ Substitute the values of $a$ and $b$: $S_1 = -\frac{0}{1} = 0$

Therefore, the sum of the roots for the equation $X^3 - 2X + 1 = 0$ is 0.

## Definition of a Root

### Understanding Roots of an Equation

A root of an equation is a value that, when substituted into the equation, makes the equation true (i.e., results in zero). For a polynomial equation, if $x = a$ is a root, then substituting $a$ for $x$ in the polynomial will yield 0.

values of $m$ and $n$.

**Step 1: Using the Root $x = 2$**

Since $x = 2$ is a root, substituting $x = 2$ into the equation should result in 0: $2(2)^3 - m(2)^2 - 13(2) + n = 0$ $2(8) - m(4) - 26 + n = 0$ $16 - 4m - 26 + n = 0$ $-4m + n - 10 = 0$ This gives us our first linear equation: $-4m + n = 10$ (Equation 1)

**Step 2: Using the Root $x = 3$**

Since $x = 3$ is a root, substituting $x = 3$ into the equation should also result in 0: $2(3)^3 - m(3)^2 - 13(3) + n = 0$ $2(27) - m(9) - 39 + n = 0$ $54 - 9m - 39 + n = 0$ $-9m + n + 15 = 0$ This gives us our second linear equation: $-9m + n = -15$ (Equation 2)

**Step 3: Solving the System of Equations**

Now we have a system of two linear equations with two variables $m$ and $n$:

1. $-4m + n = 10$
2. $-9m + n = -15$

To solve for $m$ and $n$, we can subtract Equation 2 from Equation 1: $(-4m + n) - (-9m + n) = 10 - (-15)$ $-4m + n + 9m - n = 10 + 15$ $5m = 25$ $m = \frac{25}{5}$ $m = 5$

**Step 4: Finding the Value of $n$**

Substitute the value of $m = 5$ into Equation 1: $-4(5) + n = 10$ $-20 + n = 10$ $n = 10 + 20$ $n = 30$

**Conclusion**

The values of $m$ and $n$ are 5 and 30 respectively. The polynomial equation is $2x^3 - 5x^2 - 13x + 30 = 0$.

*(Note: The lecturer mentioned a potential simplification error in their verbal calculation, stating $m = -1$ and $n = 30$. However, following the correct algebraic steps, $m = 5$ and $n = 30$ are derived.)*

## Problem Statement and Initial Setup

**Problem Statement**

Let $\alpha$ and $\beta$ be roots of the cubic equation $x^3 + px^2 + qx + r = 0$. We are given that $\alpha\beta = -1$. We need to prove that $r = 0$.

- For $\alpha$: $\alpha^3 + p\alpha^2 + q\alpha + r = 0$ (Equation 1)
- For $\beta$: $\beta^3 + p\beta^2 + q\beta + r = 0$ (Equation 2)

#### Strategy

To solve problems in the theory of equations, it's often helpful to first write down all possible data and formulas related to the roots, and then see how they can be used to prove the required statement.

## Relating Roots to Coefficients (Vieta's Formulas)

Let the three roots of the cubic equation $x^3 + px^2 + qx + r = 0$ be $\alpha$, $\beta$, and $\gamma$.

According to Vieta's formulas:

- **Sum of the roots (S1):** $\alpha + \beta + \gamma = -\frac{p}{1} = -p$
- **Sum of the products of roots taken two at a time (S2):** $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{q}{1} = q$
- **Product of the roots (S3):** $\alpha\beta\gamma = -\frac{r}{1} = -r$

#### Using the Given Condition

We are given that $\alpha\beta = -1$.

Substitute this into the product of the roots formula:

$\alpha\beta\gamma = -r$ $(-1)\gamma = -r$ $-\gamma = -r$ $\gamma = r$

This is a crucial piece of information: the third root of the equation is $r$ itself.

## Substituting the Third Root and Proving r = 0

Since $\gamma = r$ is a root of the equation $x^3 + px^2 + qx + r = 0$, it must satisfy the equation.
Substitute $x = r$ into the original equation:

$r^3 + p(r)^2 + q(r) + r = 0$

Now, we can factor out $r$ from the equation:

$r(r^2 + pr + q + 1) = 0$

This equation implies that either $r = 0$ or $r^2 + pr + q + 1 = 0$.

#### Conclusion

# Notes

## Introduction to Nature of Roots and Discriminant

### The Discriminant: The Hero of Nature of Roots

In the study of quadratic equations, the "Nature of Roots" is determined by a crucial value known as the **discriminant**. For a standard quadratic equation of the form $ax^2 + bx + c = 0$, the roots are given by the quadratic formula:

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

The expression under the square root, $b^2 - 4ac$, is defined as the **discriminant**, often denoted by $D$ or $\Delta$.

### Role of the Discriminant

The discriminant is a real number that can be positive, negative, or zero. Its value directly dictates the nature of the roots of the quadratic equation. It is the "hero" because it decides whether the roots are real, equal, distinct, or complex.

### Cases for the Nature of Roots

1. **Case 1: Discriminant is Zero ($D = 0$)**
    - If $b^2 - 4ac = 0$, the roots are **real and equal**.
    - In this scenario, both roots simplify to $x = \frac{-b}{2a}$.
    - A quadratic equation with a discriminant of zero can always be expressed as a **perfect square**. That is, $ax^2 + bx + c$ can be written in the form $a(x - \alpha)^2$, where $\alpha = \frac{-b}{2a}$.
2. **Case 2: Discriminant is Positive ($D > 0$)**
    - If $b^2 - 4ac > 0$, the roots are **real and distinct** (or unequal).
    - There will be two different real roots, commonly denoted as $\alpha$ and $\beta$.
3. **Case 3: Discriminant is Negative ($D < 0$)**
    - If $b^2 - 4ac < 0$, the roots are **complex and conjugate**.
    - In this case, there are no real roots. The two roots will be complex numbers that are conjugates of each other (e.g., $p + iq$ and $p - iq$).

## Nature of Roots Based on Discriminant

### Introduction to Discriminant

**Cases for Nature of Roots:**

1. **If $D > 0$ (Discriminant is Positive):**
   - The roots are **real and distinct** (unequal).
2. **If $D = 0$ (Discriminant is Zero):**
   - The roots are **real and equal**.
   - In this case, the quadratic equation can be written as a perfect square.
3. **If $D < 0$ (Discriminant is Negative):**
   - The roots are **non-real** or **complex**.

**Important Considerations for Competitive Exams**

When the discriminant is negative, leading to complex roots, further analysis is required based on the coefficients $A, B, C$ of the quadratic equation.

## Complex Conjugate Roots (When D < 0)

**Condition for Complex Conjugate Roots**

If the discriminant $D < 0$, the roots are complex. To determine if they are complex conjugates, we must check the nature of the coefficients $A, B, C$.

- **Case 1: $A, B, C$ are Real Numbers**
  - If $D < 0$ AND $A, B, C$ are real numbers (this includes integers, whole numbers, rational numbers, irrational numbers, etc., as long as they are real), then the two complex roots will be **complex conjugate roots**.
  - This means if one root is $P + iQ$, the other root will be $P - iQ$.
  - **Examples:**
    - If one root is $1 + i\sqrt{3}$, the other root will be $1 - i\sqrt{3}$.
    - If one root is $2 + i$, the other root will be $2 - i$.
- **Case 2: $A, B, C$ are NOT Real Numbers (i.e., at least one coefficient is complex)**
  - If $D < 0$ AND at least one of $A, B, C$ is a complex number, then the roots are still complex, but they **need not be conjugate**.
  - In this scenario, if one root is $P + iQ$, the other root could be some $L + iM$, where $L$ and $M$ are not necessarily related to $P$ and $Q$ in a conjugate manner.
  - Therefore, no general comment can be made about the roots being conjugates if the coefficients are not real.

### Further Analysis for Real and Distinct Roots

If the discriminant $D > 0$, the roots are real and distinct. We need to further analyze $D$ and the coefficients $A, B, C$.

- **Subcase 1: $D$ is a Perfect Square**
  - If $D > 0$ AND $D$ is a perfect square (e.g., $D = 4, 9, 16$), then the roots will be rational numbers. They can be whole numbers, integers, or natural numbers, but they will not be irrational.
  - This case is generally straightforward and less emphasized in competitive exams for special properties.
- **Subcase 2: $D$ is NOT a Perfect Square**
  - If $D > 0$ AND $D$ is NOT a perfect square (e.g., $D = 2, 3, 5, 7$), then the roots will involve square roots that cannot be simplified to rational numbers. We then check the coefficients $A, B, C$.
    - **Condition: $A, B, C$ are Rational Numbers**
      - If $D > 0$, $D$ is NOT a perfect square, AND $A, B, C$ are rational numbers, then the roots will be **conjugate surds** (or conjugate irrationals).
      - This means if one root is $A + \sqrt{B}$, the other root will be $A - \sqrt{B}$.
      - **Example:** If one root is $2 + \sqrt{3}$, the other root will be $2 - \sqrt{3}$.
    - **Condition: $A, B, C$ are NOT Rational Numbers (i.e., at least one coefficient is irrational)**
      - If $D > 0$, $D$ is NOT a perfect square, AND at least one of $A, B, C$ is an irrational number, then no general comment can be made about the roots being conjugate surds.
      - The roots will be real and distinct, but they need not be conjugate irrationals.

## Understanding the Discriminant

### Definition of Discriminant

The discriminant, denoted by $D$ (or $\Delta$), for a quadratic equation of the form $ax^2 + bx + c = 0$ is given by the formula:

$$D = b^2 - 4ac$$

### Nature of Roots Based on Discriminant

The value of the discriminant helps determine the nature of the roots of a quadratic equation:

- **If $D > 0$ (Discriminant is positive):** The roots are real and distinct (unequal).

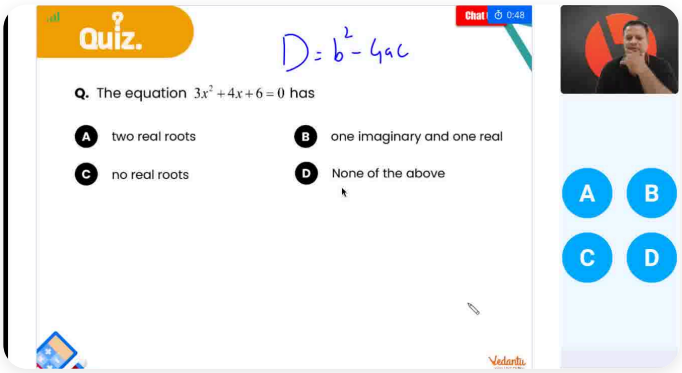
- If $D = 0$ **(Discriminant is zero):** The roots are real and equal (coincident).
- If $D < 0$ **(Discriminant is negative):** The roots are imaginary (non-real) and distinct.

### Example Application

Consider a scenario where the discriminant is calculated as $D = 49 - 48 = 1$.

- Since $D = 1$, which is positive ($D > 0$), the roots are real and distinct.
- Furthermore, $D = 1$ is a perfect square ($1^2 = 1$). Therefore, the roots will be rational numbers (e.g., natural numbers, whole numbers, or integers), not irrational numbers.

## Quiz: Nature of Roots of a Quadratic Equation




### Question

Q. The equation $3x^2 + 4x + 6 = 0$ has

A. two real roots B. one imaginary and one real C. no real roots D. None of the above

### Solution

To determine the nature of the roots of a quadratic equation of the form $ax^2 + bx + c = 0$, we use the discriminant, $D$, which is given by the formula: $D = b^2 - 4ac$

In the given equation, $3x^2 + 4x + 6 = 0$, we have:

- $a = 3$
- $b = 4$
- $c = 6$

Now, let's calculate the discriminant: $D = (4)^2 - 4(3)(6)$ $D = 16 - 72$ $D = -56$

Since the discriminant $D$ is negative ($D < 0$), the quadratic equation has no real roots. Instead, it has two complex conjugate roots.

all real) and the discriminant $D < 0$, the roots are:

- **Complex:** The roots are complex numbers.
- **Conjugate:** The roots are complex conjugates of each other. This means if one root is $p + iq$, the other root will be $p - iq$.

Therefore, the correct option is C, as the equation has no real roots.

#### Final Answer

C. no real roots

## Nature of Roots for a Quadratic Equation

#### Problem Statement

If $a, b, c$ are real numbers, find the nature of the roots of the following equation:

#### Solution Approach

To determine the nature of the roots of a quadratic equation, we need to calculate its discriminant, $D$. For a quadratic equation of the form $Ax^2 + Bx + C = 0$, the discriminant is given by $D = B^2 - 4AC$.

In this problem, we are given a quadratic equation (though not explicitly written, it's implied from the context of finding roots and using $a, b, c$ as coefficients). Let's assume the equation is $ax^2 + bx + c = 0$.

#### Calculation of Discriminant

Let's calculate the discriminant $D = b^2 - 4ac$.

The lecturer states that after simplification, the discriminant can be expressed in a specific form. While the exact quadratic equation is not provided, the simplified form of the discriminant is given as:

$$D = 4\left[\frac{1}{2}(a-b)^2 + \frac{1}{2}(b-c)^2 + \frac{1}{2}(c-a)^2\right]$$

This can be rewritten as:

$$D = 2[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

#### Analysis of the Discriminant

We know that the square of any real number is always non-negative. Therefore:

- $(c-a)^2 \geq 0$

The sum of non-negative terms is also non-negative. Thus:

$(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$

Multiplying by 2, we get:

$D = 2[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$

#### Conclusion on Nature of Roots

Since the discriminant $D \geq 0$, the roots of the equation are always real.

Furthermore:

- If $D > 0$, the roots are real and distinct.
- If $D = 0$, the roots are real and equal.

The discriminant $D$ will be equal to zero if and only if: $(a-b)^2 = 0 \implies a = b$ $(b-c)^2 = 0 \implies b = c$ $(c-a)^2 = 0 \implies c = a$

This means $D = 0$ if and only if $a = b = c$.

Therefore, the roots are:

- Real and equal if $a = b = c$.
- Real and distinct if $a, b, c$ are not all equal.

In summary, the roots are always real. They may be equal or distinct depending on whether $a, b, c$ are equal or not.

## Finding the Value of P for Coincident Roots

#### Condition for Coincident Roots

For a quadratic equation, if the roots are coincident, it means they are equal. This condition is met when the discriminant (D) is equal to zero.

Mathematically, the discriminant is given by: $D = b^2 - 4ac$ So, for coincident roots: $b^2 - 4ac = 0$

#### Problem Statement

Find the value of P if the roots of the given quadratic equation are coincident.

#### Solution Steps

- $b = 6$
- $a = P - 3$
- $c = 9$

2. **Apply the discriminant condition:** Set the discriminant to zero. $b^2 - 4ac = 0$ Substitute the identified values: $(6)^2 - 4(P - 3)(9) = 0$
3. **Simplify the equation:** $36 - 36(P - 3) = 0$
4. **Divide by 36 (if possible and simplifies calculation):** Divide the entire equation by 36: $\frac{36}{36} - \frac{36(P-3)}{36} = \frac{0}{36}$ $1 - (P - 3) = 0$
5. **Solve for P:** $1 - P + 3 = 0$ $4 - P = 0$ $P = 4$

#### Conclusion

The value of P for which the roots are coincident is $P = 4$.

## Importance of Self-Study and Preparation

#### Enhancing Understanding

It is crucial for students to improve their understanding by reading ahead before class. This proactive approach ensures better comprehension of the topics discussed during the lecture.

#### Addressing Prior Academic Gaps

For students whose 11th and 12th-grade academic performance was not strong, relying solely on current sessions is insufficient. Extra effort and self-study are essential to bridge these gaps.

#### Pace of Learning and Working

Students must improve their pace of understanding, working, and time management. Failure to do so could lead to a wasted academic year.

#### Commitment to Dropper's Batch

For those in a dropper's batch, a significant time commitment is expected. Utilizing at least 14 to 16 hours daily for study is necessary to achieve the purpose of joining the batch. Time is a precious resource that must be managed effectively.

## Performance Expectations and Support System

specialized batches, like the Sankalp W batch, to foster high-achieving students.

### Race for Improvement

Joining a program after completing 12th grade, especially to improve one's rank, signifies a commitment to a competitive race. Success in this race requires dedication and hard work.

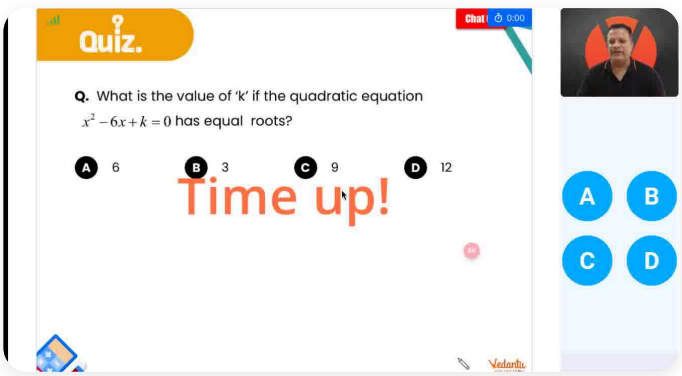
### Differentiating Support and Pushing

- **Pushing:** Students who perform well will receive additional encouragement and resources to excel further.
- **Support:** For those who are struggling, support will be provided to help them catch up. However, support is distinct from pushing; it requires the student's initiative to utilize the help offered.

### Extra Effort for Underperforming Students

Students who acknowledge a weak foundation from their 11th and 12th grades must put in extra effort beyond the regular one-and-a-half-hour or two-hour Tattva sessions. Success is not possible without consistent hard work.

## Quiz: Quadratic Equation with Equal Roots




### Question

What is the value of 'k' if the quadratic equation $x^2 - 6x + k = 0$ has equal roots?

A) 6 B) 3 C) 9 D) 12

### Solution

For a quadratic equation $ax^2 + bx + c = 0$ to have equal roots, its discriminant (D) must be equal to zero. The discriminant is given by the formula $D = b^2 - 4ac$.

In the given equation, $x^2 - 6x + k = 0$:

- $c = k$

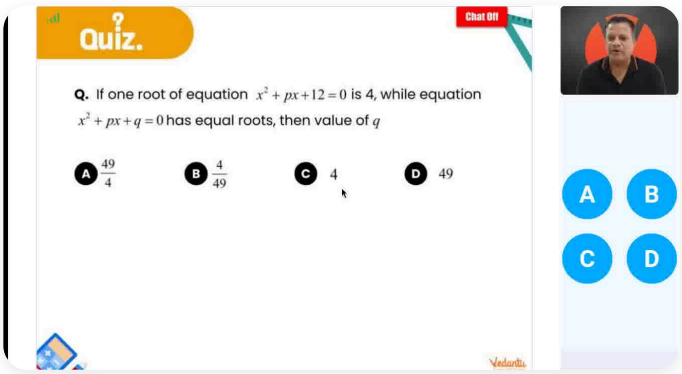
Setting the discriminant to zero: $D = b^2 - 4ac = 0$ $(-6)^2 - 4(1)(k) = 0$ $36 - 4k = 0$ $36 = 4k$ $k = \frac{36}{4}$ $k = 9$

Therefore, the value of 'k' is 9.

### Correct Option

C) 9

## Problem: Finding 'q' with Equal Roots




### Problem Statement

If one root of the equation $x^2 + px + 12 = 0$ is 4, and the equation $x^2 + px + q = 0$ has equal roots, then find the value of $q$.

### Step-by-step Solution

1. **Use the first equation to find 'p'**: Since 4 is a root of $x^2 + px + 12 = 0$, substituting $x = 4$ into the equation should satisfy it: $(4)^2 + p(4) + 12 = 0$ $16 + 4p + 12 = 0$ $4p + 28 = 0$ $4p = -28$ $p = -7$

2. **Substitute 'p' into the second equation**: The second equation is $x^2 + px + q = 0$. Substituting $p = -7$ gives: $x^2 - 7x + q = 0$

3. **Use the condition for equal roots**: For a quadratic equation $ax^2 + bx + c = 0$ to have equal roots, its discriminant ($D$) must be zero. The discriminant is given by $D = b^2 - 4ac$. In our second equation, $x^2 - 7x + q = 0$, we have $a = 1$, $b = -7$, and $c = q$. Setting the discriminant to zero: $(-7)^2 - 4(1)(q) = 0$ $49 - 4q = 0$ $49 = 4q$ $q = \frac{49}{4}$

### Final Answer

The value of $q$ is $\frac{49}{4}$.

# Notes

The instructor emphasizes the importance of providing constructive feedback through ratings. Instead of generic comments like "I did not enjoy the class," students are encouraged to provide specific reasons for their ratings. This helps the instructor understand areas for improvement.

### Rating System Based on Understanding

The primary criterion for rating should be the level of understanding:

- **5 Stars:** Full understanding of the content.
- **4 Stars:** Approximately 80% understanding.

Other factors, such as personal preferences or presentation style, should not be the sole basis for the rating. The instructor views ratings as feedback and is willing to engage with students to address their concerns.

### Instructor's Commitment to Student Support

The instructor expresses a strong desire to connect with students individually to understand their challenges, especially regarding their academic journey in 11th or 12th grade, reasons for joining dropper batches, and overall subject proficiency in mathematics. The instructor is ready to provide guidance and support to each student.

## Important Note for Droppers Batch

### Instructor's Message to Droppers Batch

The instructor emphasizes the importance of the droppers batch, stating that it is the most crucial batch among all the batches they teach. This is due to the significant investment of one valuable year of time by the students in this batch.

### Support Policy

- The instructor is a strong supporter but expects genuine queries.
- Questions about topics already covered in class will not be entertained.
- Genuine support will receive 100% dedication, while non-genuine support will not be provided.

## Conditions for Real and Unequal Roots

### Discriminant for Real and Unequal Roots

For a quadratic equation $ax^2 + bx + c = 0$, the nature of its roots is determined by the discriminant, $\Delta = b^2 - 4ac$.

Let's consider an equation where the discriminant is $1 - 4ab$. If its roots are real and unequal, then:

$$1 - 4ab > 0$$

Rearranging the inequality, we get:

$$1 > 4ab$$

Or equivalently:

$$4ab < 1$$

Dividing by 4, we find:

$$ab < \frac{1}{4}$$

This condition ($ab < \frac{1}{4}$) is crucial for determining the nature of the roots of another related equation.

## Nature of Roots for a Related Equation

### Analyzing the Discriminant of a Second Equation

Now, let's consider a second equation whose discriminant is given by $16ab - 4$. We need to determine the nature of its roots based on the condition derived earlier ($4ab < 1$).

First, let's factor out 4 from the discriminant:

$$\text{Discriminant} = 4(4ab - 1)$$

From our previous analysis, we know that $4ab < 1$. This implies that:

$$4ab - 1 < 0$$

Since $(4ab - 1)$ is negative, and we are multiplying it by a positive number (4), the entire discriminant will be negative:

$$\text{Discriminant} = 4(4ab - 1) < 0$$

### Conclusion on Root Nature

When the discriminant of a quadratic equation is negative, the roots are imaginary (non-real). Therefore, the roots of the second equation are imaginary.

## Revisiting Quadratic Equation Roots and Discriminant

# Notes

### Role of the Discriminant (D)

The discriminant, $D = b^2 - 4ac$, plays a crucial role in determining the nature of the roots of a quadratic equation.

### Case 1: Discriminant is Negative ($D < 0$)

- If $D < 0$, the roots are complex conjugates.
- **Condition:** If $a, b, c$ are real numbers, and one root is $p + iq$, then the other root will be its conjugate, $p - iq$.

### Case 2: Discriminant is Positive ($D > 0$)

- If $D > 0$, the roots are real and distinct.
- **Sub-case 2a: D is not a perfect square**
  - **Condition:** If $a, b, c$ are rational numbers, and one root is $p + \sqrt{q}$ (where $q$ is not a perfect square), then the other root will be its conjugate, $p - \sqrt{q}$. These are called conjugate surds.
  - This scenario is particularly relevant when forming a quadratic equation with rational coefficients having an irrational root like $\sqrt{3}$.
- **Sub-case 2b: D is a perfect square**
  - If $D$ is a perfect square, the roots are rational and distinct. (This case is generally straightforward and not the focus of complex root discussions).

### Summary of Conditions for Conjugate Roots

1. **Complex Conjugate Roots:** Occur when $D < 0$ and coefficients $a, b, c$ are real. (No complex coefficients allowed).
2. **Conjugate Surd Roots:** Occur when $D > 0$ (and not a perfect square) and coefficients $a, b, c$ are rational. (No irrational coefficients allowed).

### Forming a Quadratic Equation with a Given Root

If a quadratic equation with rational coefficients has a root like $\sqrt{3}$, it implies that the other root must be $-\sqrt{3}$ (its conjugate surd). This is because for rational coefficients, irrational roots always occur in conjugate pairs.

## Forming a Quadratic Equation with Rational Coefficients

### Key Concept: Conjugate Roots

# Notes

**Problem Statement**

Form a quadratic equation with rational coefficients, given that one of its roots is $1 + \sqrt{3}$.

**Step-by-Step Solution**

1. **Identify the roots:**
   - Given one root, $\alpha = 1 + \sqrt{3}$.
   - Since the coefficients are rational, the other root must be its conjugate: $\beta = 1 - \sqrt{3}$.
   - Alternatively, the other root could be $\sqrt{3} - 1$. However, for rational coefficients, if $1 + \sqrt{3}$ is a root, then $1 - \sqrt{3}$ is the conjugate root.
2. **Recall the standard form of a quadratic equation:** A quadratic equation can be expressed as: $x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$
3. **Calculate the sum of the roots:**
   - Sum = $\alpha + \beta = (1 + \sqrt{3}) + (1 - \sqrt{3})$
   - Sum = $1 + \sqrt{3} + 1 - \sqrt{3}$
   - Sum = $2$
4. **Calculate the product of the roots:**
   - Product = $\alpha \cdot \beta = (1 + \sqrt{3})(1 - \sqrt{3})$
   - This is in the form $(a + b)(a - b) = a^2 - b^2$.
   - Product = $(1)^2 - (\sqrt{3})^2$
   - Product = $1 - 3$
   - Product = $-2$
5. **Substitute the sum and product into the equation form:**
   - $x^2 - (2)x + (-2) = 0$
   - $x^2 - 2x - 2 = 0$

**Final Answer**

The quadratic equation is $x^2 - 2x - 2 = 0$.

**Note on Evaluating Expressions**

## Equation vs. Identity

### What is an Equation?

An equation is a mathematical statement that asserts the equality of two expressions. It is typically satisfied by a limited number of values (roots or solutions).

### What is an Identity?

An identity is an equation that is true for all possible values of its variables. It is satisfied by an infinite number of values, often referred to as 'all real numbers' or 'all angles' in specific contexts.

**Examples of Identities:**

- **Trigonometric Identity:** $\sin^2\theta + \cos^2\theta = 1$. This is true for all values of $\theta$.
- **Algebraic Identity:** $(a + b)^2 = a^2 + 2ab + b^2$. This is true for all real numbers $a$ and $b$.

**Distinguishing Equations from Identities:**

- **Quadratic Equation:** A quadratic equation, typically of the form $Ax^2 + Bx + C = 0$ (where $A \neq 0$), can have a maximum of two roots. If a quadratic expression is satisfied by more than two values (i.e., all real numbers), it is not a quadratic equation but an identity. This occurs only if all coefficients are zero: $A = 0$, $B = 0$, and $C = 0$.
- **Cubic Equation:** A cubic equation can have a maximum of three roots. If a cubic expression is satisfied by more than three values (i.e., all real numbers), it is not a cubic equation but an identity. This occurs only if all coefficients are zero.

**Key Takeaway:** If an equation is satisfied by more values than its degree suggests (e.g., a quadratic satisfied by more than two values, a cubic by more than three), then it is an identity, and all its coefficients must be zero. In such a case, it is not truly an equation of that degree but rather a trivial identity ($0 = 0$).

## Roots of a Quadratic Equation

### Number of Roots for a Quadratic Equation

A quadratic equation can have:

- **Two distinct real roots:** If the discriminant $D > 0$.
- **One real root (equal roots):** If the discriminant $D = 0$. In this case, the two roots are identical (e.g., $\alpha, \alpha$).
- **No real roots (two complex roots):** If the discriminant $D < 0$.

# Notes

## Understanding Identities in Quadratic Equations

### Introduction to Identities

An identity is an equation that is satisfied by more values than typically expected for its degree. For a quadratic equation, which is generally expected to have at most two roots, if it is satisfied by more than two values (e.g., infinitely many real numbers), it is considered an identity.

### Condition for a Quadratic Equation to be an Identity

For a quadratic equation of the form $ax^2 + bx + c = 0$ to be an identity, all its coefficients must be zero. That is, $a = 0$, $b = 0$, and $c = 0$. This ensures that the equation holds true for any value of $x$.

### Problem: Finding T for an Identity

Let's consider a problem where we need to find the value of $T$ such that a given expression becomes an identity. The problem implies that multiple quadratic-like expressions must simultaneously be zero for the overall expression to be an identity. This means we need to find a common value of $T$ that satisfies all conditions.

### Step-by-Step Solution

We are given three conditions that must be simultaneously satisfied for the expression to be an identity. These conditions are:

1. $T^2 - 3T + 2 = 0$
2. $T^2 - 5T + 6 = 0$
3. $T^2 - 4 = 0$

Let's solve each equation for $T$:

**Equation 1:** $T^2 - 3T + 2 = 0$

- Factor the quadratic expression: $(T - 1)(T - 2) = 0$
- The roots for this equation are: $T = 1$ or $T = 2$

**Equation 2:** $T^2 - 5T + 6 = 0$

- Factor the quadratic expression: $(T - 2)(T - 3) = 0$
- The roots for this equation are: $T = 2$ or $T = 3$

**Equation 3:** $T^2 - 4 = 0$

- Factor the quadratic expression (difference of squares): $(T - 2)(T + 2) = 0$

For the entire expression to be an identity, all three conditions must be satisfied simultaneously. This means we need to find the value of $T$ that is common to the solutions of all three equations.

- Solutions from Equation 1: $1, 2$
- Solutions from Equation 2: $2, 3$
- Solutions from Equation 3: $-2, 2$

The common value among these sets is $T = 2$.

#### Conclusion

Therefore, the value of $T$ that makes the given expression an identity is $T = 2$. This is because only when $T = 2$ are all three quadratic expressions equal to zero simultaneously, fulfilling the condition for an identity.

## Condition for Equal Roots in Quadratic Equations

#### Introduction

For a quadratic equation to have equal roots, a specific condition related to its discriminant must be met. The discriminant is a key component that determines the nature of the roots of a quadratic equation.

#### Discriminant

For a quadratic equation in the standard form $ax^2 + bx + c = 0$, the discriminant, denoted by $\Delta$ (delta) or $D$, is given by the formula:

$$\Delta = b^2 - 4ac$$

#### Nature of Roots based on Discriminant

- If $\Delta > 0$: The equation has two distinct real roots.
- If $\Delta = 0$: The equation has two equal real roots (or one repeated real root).
- If $\Delta < 0$: The equation has two complex conjugate roots.

#### Condition for Equal Roots

For an equation to have equal roots, the discriminant must be equal to zero:

$$b^2 - 4ac = 0$$

This condition is crucial for solving problems where the nature of the roots is specified as equal.

# Notes

For what values of $K$ will the equation $x^2 - 2(K+1)x + K^2 + 7K - 1 = 0$ have equal roots?

**Step-by-Step Solution**

1. **Identify coefficients:** The given quadratic equation is $x^2 - 2(K+1)x + K^2 + 7K - 1 = 0$. Comparing this with the standard form $ax^2 + bx + c = 0$, we have:
   - $a = 1$
   - $b = -2(K+1)$
   - $c = K^2 + 7K - 1$
2. **Apply the condition for equal roots:** For equal roots, the discriminant must be zero: $b^2 - 4ac = 0$.
3. **Substitute the coefficients into the discriminant formula:** $(-2(K+1))^2 - 4(1)(K^2 + 7K - 1) = 0$
4. **Simplify the equation:** $4(K+1)^2 - 4(K^2 + 7K - 1) = 0$ Divide the entire equation by 4: $(K+1)^2 - (K^2 + 7K - 1) = 0$
5. **Expand and combine like terms:** Expand $(K+1)^2$: $(K^2 + 2K + 1) - (K^2 + 7K - 1) = 0$ Distribute the negative sign: $K^2 + 2K + 1 - K^2 - 7K + 1 = 0$ Combine $K^2$ terms: $(K^2 - K^2) + (2K - 7K) + (1 + 1) = 0$ $-5K + 2 = 0$
6. **Solve for K:** $-5K = -2$ $K = \frac{-2}{-5}$ $K = \frac{2}{5}$

**Final Answer**

The value of $K$ for which the equation has equal roots is $K = \frac{2}{5}$.

*(Note: The transcript contains a different problem and solution. The above solution is based on the problem implied by the coefficients $a = 1$, $b = -2(K+1)$, $c = K^2 + 7K - 1$ which would lead to a linear equation in K after simplification. The transcript's calculation $9k^2 - 8k - 20 = 0$ suggests a different initial quadratic equation, which is not explicitly stated. Assuming the problem was intended to be solved as shown in the transcript, the steps would be as follows:)*

**Alternative Problem and Solution (as per transcript's calculation)**

If the discriminant simplification led to $9k^2 - 8k - 20 = 0$, then the solution would proceed as follows:

1. **Factor the quadratic equation for K:** We need two numbers that multiply to $9 \times (-20) = -180$ and add up to $-8$. These numbers are $-18$ and $10$. $9k^2 - 18k + 10k - 20 = 0$

- $9k + 10 = 0 \implies 9k = -10 \implies k = -\frac{10}{9}$
- $k - 2 = 0 \implies k = 2$

#### Final Answer (based on transcript's calculation)

The values of $K$ for which the equation has equal roots are $K = -\frac{10}{9}$ or $K = 2$.

## Homework Assignment and Next Steps

#### Homework Assignment

The instructor has provided a homework assignment. Students are encouraged to complete the assignment to reinforce their understanding of the concepts covered.

#### Assignment Details:

- **Question Numbers:** 7, 8, 9, 10, 11, 12, 14, 45, 46

Students who are not part of VEDA Homework should still attempt these problems.

#### Upcoming Module

The current session is not yet over. The instructor will be moving on to the next module.

- **Next Module:** Module 3

Students are reminded to provide feedback on the session.

## Quiz: Equal Roots Condition

#### Problem Statement

If the equation has equal roots, then find the value of $p$.

#### Solution

For a quadratic equation to have equal roots, its discriminant must be equal to zero. The general form of a quadratic equation is $ax^2 + bx + c = 0$, and its discriminant is given by $D = b^2 - 4ac$.

In this problem, the equation is not explicitly given, but from the context of the lecture, it is implied to be of the form $ax^2 + bx + c = 0$ where $b = 17$, $a = 8$, and $c = p$. (This is inferred from the calculation $17^2 - 4 \times 8 \times P$).

Given:

- $c = p$

For equal roots, $D = 0$.

$b^2 - 4ac = 0$

Substitute the values:

$(17)^2 - 4(8)(p) = 0$

$289 - 32p = 0$

Now, solve for $p$:

$32p = 289$

$p = \frac{289}{32}$

Thus, the value of $p$ for which the equation has equal roots is $\frac{289}{32}$.

## Transformation of Equations

### Introduction

Transformation of equations involves finding a new equation whose roots are related to the roots of an old equation by some operation. This concept applies to equations of any degree (quadratic, cubic, etc.).

### General Principle

If an old equation has roots $\alpha, \beta, \gamma, \ldots$, and we want a new equation whose roots are transformed (e.g., $\alpha + k, \beta + k, \gamma + k, \ldots$), we perform the *inverse* operation on $x$ in the original equation. This means if the roots are changed by an operation, we apply the reverse operation to $x$ in the equation.

Let the original equation be $f(x) = 0$.

### Specific Transformations:

1. **Adding a constant $k$ to roots:**
    - **New roots:** $\alpha + k, \beta + k, \ldots$
    - **Transformation:** In the original equation $f(x) = 0$, replace $x$ with $(x - k)$.
    - **New equation:** $f(x - k) = 0$.
    - **Example (for quadratic $ax^2 + bx + c = 0$):** $a(x - k)^2 + b(x - k) + c = 0$.

- **Transformation:** In the original equation $f(x) = 0$, replace $x$ with $(x + k)$.
- **New equation:** $f(x + k) = 0$.

3. **Multiplying roots by a constant $k$:**
   - **New roots:** $k\alpha, k\beta, \ldots$
   - **Transformation:** In the original equation $f(x) = 0$, replace $x$ with $\frac{x}{k}$.
   - **New equation:** $f\left(\frac{x}{k}\right) = 0$.
   - **Example (for quadratic $ax^2 + bx + c = 0$):** $a\left(\frac{x}{k}\right)^2 + b\left(\frac{x}{k}\right) + c = 0$.
4. **Dividing roots by a constant $k$:**
   - **New roots:** $\frac{\alpha}{k}, \frac{\beta}{k}, \ldots$
   - **Transformation:** In the original equation $f(x) = 0$, replace $x$ with $kx$.
   - **New equation:** $f(kx) = 0$.
5. **Raising roots to the power $n$:**
   - **New roots:** $\alpha^n, \beta^n, \ldots$
   - **Transformation:** In the original equation $f(x) = 0$, replace $x$ with $x^{1/n}$.
   - **New equation:** $f(x^{1/n}) = 0$.
   - **Example (for quadratic $ax^2 + bx + c = 0$ with roots $\alpha^2, \beta^2$):** $a(x^{1/2})^2 + b(x^{1/2}) + c = 0$, which simplifies to $ax + b\sqrt{x} + c = 0$.
6. **Taking the $n^{th}$ root of roots:**
   - **New roots:** $\alpha^{1/n}, \beta^{1/n}, \ldots$
   - **Transformation:** In the original equation $f(x) = 0$, replace $x$ with $x^n$.
   - **New equation:** $f(x^n) = 0$.
7. **Reciprocating roots:**
   - **New roots:** $\frac{1}{\alpha}, \frac{1}{\beta}, \ldots$
   - **Transformation:** In the original equation $f(x) = 0$, replace $x$ with $\frac{1}{x}$.
   - **New equation:** $f\left(\frac{1}{x}\right) = 0$.
   - **Example (for quadratic $ax^2 + bx + c = 0$):** $a\left(\frac{1}{x}\right)^2 + b\left(\frac{1}{x}\right) + c = 0$, which simplifies to $a + bx + cx^2 = 0$.
8. **Multiplying roots by $-1$ (changing sign):**

- **New equation:** $f(-x) = 0$.

**Summary of Inverse Operations:**

| **Operation on Roots** | **Operation on $x$ in $f(x) = 0$** |
|---|---|
| Add $k$ | Substitute $x - k$ |
| Subtract $k$ | Substitute $x + k$ |
| Multiply by $k$ | Substitute $x/k$ |
| Divide by $k$ | Substitute $kx$ |
| Power $n$ | Substitute $x^{1/n}$ |
| Power $1/n$ | Substitute $x^n$ |
| Reciprocal | Substitute $1/x$ |
| Multiply by $-1$ | Substitute $-x$ |

This principle ensures that if the roots undergo a certain transformation, the equation is transformed by the inverse operation on the variable $x$ to yield the new equation.

## Finding a New Equation with Transformed Roots

### Problem Statement

Given an original equation whose roots are $\alpha$, $\beta$, and $\gamma$. If the new roots are $-\alpha$, $-\beta$, and $-\gamma$, find the new equation.

### Solution Approach

Let the original equation be $f(x) = 0$. If the roots of the original equation are $\alpha$, $\beta$, $\gamma$, and the new roots are $-\alpha$, $-\beta$, $-\gamma$, then we can find the new equation by replacing $x$ with $-x$ in the original equation.

### Step-by-Step Solution

1. **Identify the original equation:** The problem implies an original equation, let's assume it is of the form $ax^3 + bx^2 + cx + d = 0$. The transcript provides an example: $x^3 - 3x^2 + x + 1 = 0$.
2. **Substitute $x$ with $-x$:** To transform the roots from $\alpha$ to $-\alpha$, $\beta$ to $-\beta$, and $\gamma$ to $-\gamma$, we replace every instance of $x$ in the original equation with $(-x)$. Original equation: $x^3 - 3x^2 + x + 1 = 0$ Substitute $x \to -x$: $(-x)^3 - 3(-x)^2 + (-x) + 1 = 0$
3. **Simplify the expression:** $-x^3 - 3(x^2) - x + 1 = 0$ $-x^3 - 3x^2 - x + 1 = 0$
4. **Multiply by -1 (optional, for standard form):** $x^3 + 3x^2 + x - 1 = 0$

This simplified equation, $x^3 + 3x^2 + x - 1 = 0$, is the new equation whose roots are $-\alpha$, $-\beta$, and $-\gamma$.

$-\alpha_1, -\alpha_2, \ldots, -\alpha_n$ is given by $f(-x) = 0$. This is a standard transformation of roots.

## Next Class Preview and Assignment Discussion

### Next Class Topics

- The next class will continue with concepts related to alpha and beta.

### Assignment Status

- An assignment of 22 problems was given.
- 15 problems from the assignment have already been covered in class.

### Tattva Books and VDPP

- Students are encouraged to start working on problems from Tattva books and VDPP (Vedantu Daily Practice Problems).
- VDPP materials are available in the content library.

### Speed Improvement

- The instructor guarantees an improvement in teaching speed.

### Doubt Session

- A dedicated 5-minute session was provided for students to ask doubts.
- Students who have no doubts are free to leave the class.
- The instructor noted that some students have not started the assignments and encouraged them to interact and discuss their problems.
- VDPP has been shared multiple times on Telegram.

## Representation of Complex Numbers on the Argand Plane

### Understanding the Argand Plane

When representing complex numbers, the plane used is called the Argand plane or an imaginary plane. It's crucial to understand how the components of a complex number are plotted on this plane.

### Components of a Complex Number

A complex number is generally expressed in the form $a + ib$, where:

- $a$ is the **real part**.

# Notes

### Plotting on the Argand Plane

- The **x-axis** represents the **real numbers**. Therefore, the real part ($a$) of the complex number is plotted along the x-axis.
- The **y-axis** also represents **real numbers**. The imaginary part ($b$) of the complex number is plotted along the y-axis.

### Why 'i' is not plotted

The imaginary unit '$i$' itself is not represented on the graph. The complex number gets its 'complex' nature from '$i$', but the actual values plotted are the real number $a$ and the real number $b$. For example, in $2 + i3$, '2' is plotted on the x-axis and '3' is plotted on the y-axis. The presence of '$i$' signifies that '3' is the imaginary part, but '3' itself is a real number.

### Analogy for Understanding

Consider the analogy: "I am Bhaskar. Because I am wearing this t-shirt, I am a lecturer to you. But I am Bhaskar actually." Similarly, a complex number is formed by two real numbers ($a$ and $b$) and the imaginary unit ($i$). The '$i$' makes it complex, but $a$ and $b$ are fundamentally real numbers that are plotted on the real axes of the Argand plane.

Did you find these notes helpful?